

**PCB Design**

**PCB DESIGN HAND BOOK**

## Introduction

This document is aimed for those who are new to PCB design. It highlights and explains PCB design and construction. None of this material is new. On the contrary it has been collected from different application notes and books and interestingly much of this material is not covered in college curricula.

A **printed circuit board**, or **PCB**, is used to mechanically support and electrically connect electronic components using conductive pathways, or traces, etched from copper sheets laminated onto a non-conductive *substrate*. Alternative names are **printed wiring board (PWB)**, and **etched wiring board**. A PCB populated with electronic components is a **printed circuit assembly (PCA)**, also known as a **printed circuit board assembly (PCBA)**.

PCBs are rugged, inexpensive, and can be highly reliable. They require much more layout effort and higher initial cost than either wire-wrapped or point-to-point constructed circuits, but are much cheaper and faster for high-volume production.

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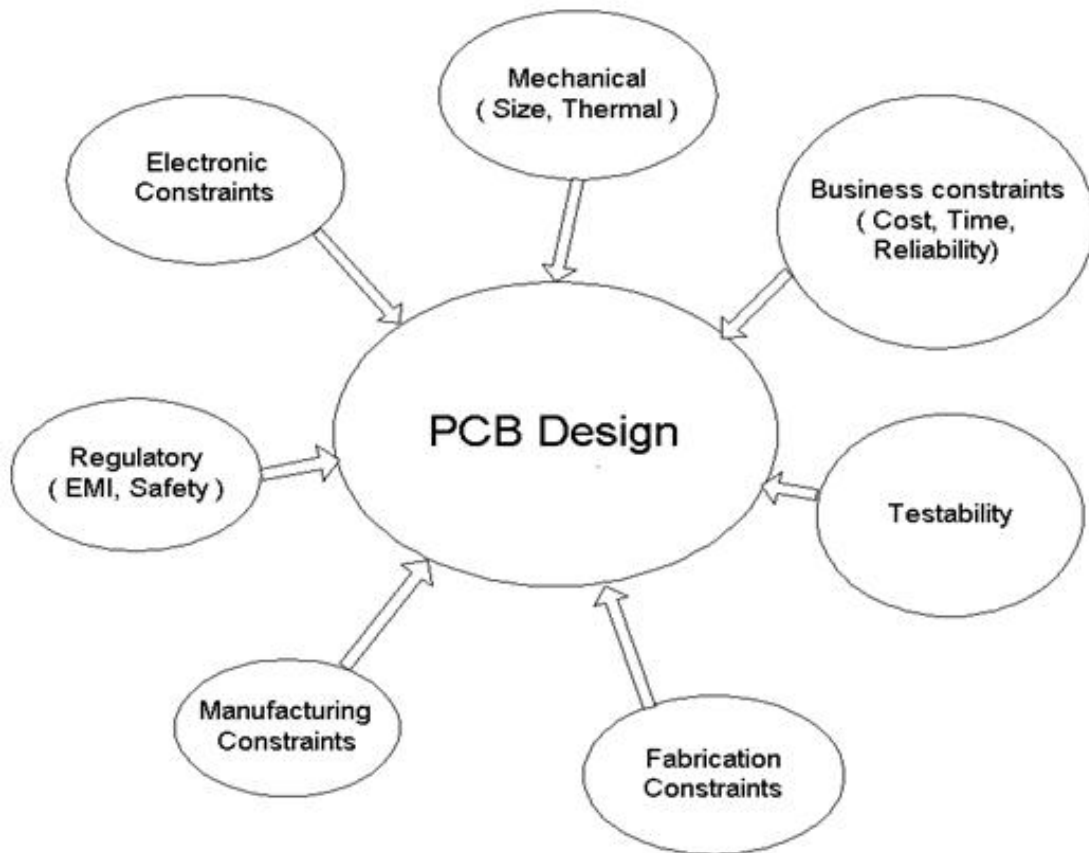
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## Interdependencies of PCB design



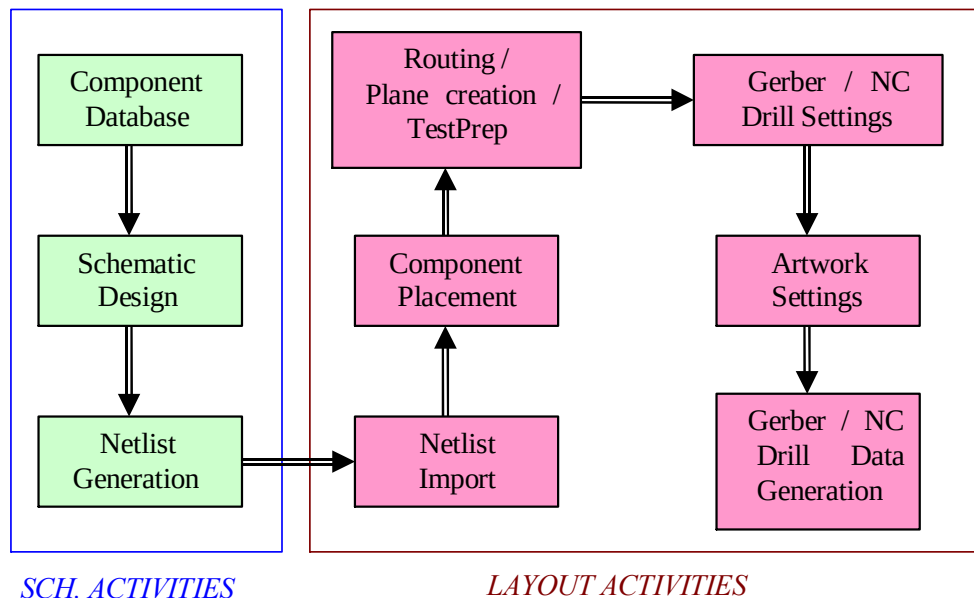
## Industry leading PCB design tools and vendors

| Vendors                | Tools                           |
|------------------------|---------------------------------|
| Cadence Design systems | Allegro, OrCad Layout           |
| Mentor Graphics        | PADS, Expedition, Board station |
| Altium                 | Altium designer, P-CAD          |
| Zuken                  | Visula, Cadstar                 |

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## PCB Design Process Flow



## Standards

There are industry standards for almost every aspect of PCB design. These standards are controlled by the former Institute for Interconnecting and Packaging Electronic Circuits, who are now known simply as the IPC ([www.ipc.org](http://www.ipc.org)). There is an IPC standard for every aspect of PCB design, manufacture, testing, and anything else that you could ever need. The major document that covers PCB design is IPC-2221, “Generic Standard on Printed Board Design”.

This standard superseded the old IPC-D-275 standard (also Military Std 275) which has been used for the last half century. Local countries also have their own various standards for many aspects of PCB design and manufacture, but by and large the IPC standards are the accepted industry standard around the world. Printed Circuit Boards are also known (some would say, more correctly known) as Printed Wiring Boards, or simply Printed Boards. But we will settle on the more common term PCB for this article.

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## Performance Classes

Three general end product classes have been established to reflect progressive increases in sophistication, functional performance requirements and testing inspection frequency. It should be recognized that there might be an overlap of equipment between classes.

***Class 1 General Electronic Products*** - Includes consumer products, some computer and computer peripherals, as well as general military hardware suitable for applications where cosmetic imperfections are not important and the major requirement is function of the completed printed board or printed board assembly.

***Class 2 Dedicated Service Electronic Products*** - Includes communications equipment, sophisticated business machines, instruments and military equipment where high performance and extended life is required, and for which uninterrupted service is desired but is not critical. Certain cosmetic imperfections are allowed.

***Class 3 High Reliability Electronic Products*** - Includes the equipment for commercial and military products where continued performance or performance on demand is critical.

## The Schematic

Before you even begin to lay out your PCB, you **MUST** have a complete and accurate schematic diagram. Many people jump straight into the PCB design with nothing more than the circuit in their head, or the schematic drawn on loose post-it notes with no pin numbers and no order. This just isn't good enough, if you don't have an accurate schematic then your PCB will most likely end up a mess, and take you twice as long as it should.

A PCB design is a manufactured version of your schematic, so it is natural for the PCB design to be influenced by the original schematic. If your schematic is neat, logical and clearly laid out, then it really does make your PCB design job a lot easier. Good practice will have signals flowing from inputs at the left to outputs on the right. With electrically important sections drawn correctly, the way the designer would like them to be laid out on the PCB. Like putting bypass capacitors next to the component they are meant for. Little notes on the schematic that aid in the layout are very useful. For instance, "this pin requires a guard track to signal ground", makes it clear to the person laying out the board what precautions must be taken. Even if it is you who designed the circuit and drew the schematic, notes not only remind yourself when it comes to laying out the board, but they are useful for people reviewing the design.

## Netlists

A netlist is essentially a list of connections ("nets") which correspond to your schematic. It also contains the list of components, component designators, component footprints and other information related to your schematic. The netlist file can be

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generated by your schematic package. Generating a netlist is also called “schematic capture”.

Your PCB package can then import this netlist file and do many things. It can automatically load all the required components onto your blank board. It can also assign a “net” name to each of your component pins. With nets assigned to your PCB components, it is now possible to Auto Route, do Design Rule Checking, and display component connectivity. This is the fundamental concept behind modern Schematic and PCB CAD packages.

### Imperial and Metric

The first thing to know about PCB design is what measurement units are used and their common terminologies, as they can be awfully confusing! As any long time PCB designer will tell you, you should always use imperial units (i.e. inches) when designing PCBs. This isn't just for the sake of nostalgia, although that is a major reason! The majority of electronic components were (and still are) manufactured with imperial pin spacing. So this is no time to get stubborn and refuse to use anything but metric units, metric will make laying out of your board a lot harder and a lot messier. If you are young enough to have been raised in the metric age then you had better start learning what inches are all about and how to convert them. An old saying for PCB design is “thou shall use thous”. A tad confusing until you know what a “thou” is. A “thou” is 1/1000<sup>th</sup> of an inch, and is universally used and recognised by PCB designers and manufacturers everywhere. So start practicing speaking in terms of “10 thou spacing” and “25 thou grid”, you'll sound like a professional in no time!

Now that you understand what a thou is, we'll throw another spanner in the works with the term “mil” (or “mils”). 1 “mil” is the same as 1 thou, and is NOT to be confused with the millimeter (mm), which is often spoken the same as “mil”. The term “mil” comes from 1 thou being equal to 1 mili inch. As a general rule avoid the use of “mil” and stick to “thou”, it's less confusing when trying to explain PCB dimensions to those metricated non-PCB people.

Some PCB designers will tell you not to use metric millimeters for ANYTHING to do with a PCB design. In the practical world though, you'll have to use both imperial inches (thous) and the metric millimeter (mm). So which units do you use for what? As a general rule, use thous for tracks, pads, spacings and grids, which are most of your basic “design and layout” requirements. Only use mm for “mechanical and manufacturing” type requirements like hole sizes and board dimensions.

You will find that many PCB manufacturers will follow these basic guidelines also, for when they ask you to provide details for a quote to manufacture your board. Most manufacturers use metric size drills, so specifying imperial size holes really is counterproductive and can be prone to errors. Just to confuse the issue even further, there are many components (new surface mount parts are an example) which have metric pin spacing and dimensions. So you'll often have to design some component footprints using metric grids and pads. Many component datasheets will also have metric dimensions even though the spacing are designed to an imperial grid. If you see a “weird” metric dimension like 1.27mm in a component, you can be pretty sure it



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actually has a nice round imperial equivalent. In this case 1.27mm is 50 thou. Yes, PCB design can be confusing!

So whatever it is you have to do in PCB design you'll need to become an expert at imperial to metric conversion, and vice-versa. It will help you greatly if you

memorise a few key conversions, like **100 thou (0.1 inch) = 2.54mm, and 200 thou (0.2 inch) = 5.08mm** etc. Values of 100 thou and above are very often expressed in inches instead of thous. So 0.2 inch is more commonly used than 200thou. 1 inch is also commonly known as 1 "pitch". So it is common to hear the phrase "0.1 inch pitch", or more simply "0.1 pitch" with the inches units being assumed. This is often used for pin spacing on components. 100 thou is a basic "reference point" for all aspects of PCB design, and a vast array of common component lead spacing are multiples or fractions of this basic unit. 50 and 200 thou are the most common. Along with the rest of the world, the IPC standards have all been metricated, and only occasionally refer to imperial units. This hasn't really converted the PCB industry though. Old habits die hard, and imperial still reigns supreme in many areas of practical usage.

### Working to Grids

The second major rule of PCB design, and the one most often missed by beginners, is to lay out your board on a fixed grid. This is called a "snap grid", as your cursor, components and tracks will "snap" into fixed grid positions. Not just any size grid mind you, but a fairly coarse one. 100 thou is a standard placement grid for very basic through hole work, with 50 thou being a standard for general tracking work, like running tracks between through hole pads. For even finer work you may use a 25 thou snap grid or even lower. Many designers will argue over the merits of a 20 thou grid vs a 25 thou grid for instance. In practice, 25 thou is often more useful as it allows you to go exactly half way between 50 thou spaced pads. Why is a coarse snap grid so important? It's important because it will keep your components neat and symmetrical; aesthetically pleasing if you may. It's not just for aesthetics though - it makes future editing, dragging, movement and alignment of your tracks, components and blocks of components easier as your layout grows in size and complexity. A bad and amateurish PCB design is instantly recognisable, as many of the tracks will not line up exactly in the center of pads. Little bits of tracks will be "tacked" on to fill in gaps etc. This is the result of not using a snap grid effectively.

Good PCB layout practice would involve you starting out with a coarse grid like 50 thou and using a progressively finer snap grid if your design becomes "tight" on space. Drop to 25 thou and 10 thou for finer routing and placement when needed. This will do 99% of boards. Make sure the finer grid you choose is a nice even division of your standard 100 thou. This means 50, 25, 20, 10, or 5 thou. Don't use anything else, you'll regret it.

A good PCB package will have hotkeys or programmable macro keys to help you switch between different snap grid sizes instantly, as you will need to do this often.

There are two types of grids in a PCB drafting package, a snap grid as discussed, and a "visible" grid. The visible grid is an optional on-screen grid of solid or dashed lines,

or dots. This is displayed as a background behind your design and helps you greatly in lining up components and tracks. You can have the snap grid and visible grid set

to different units (metric or imperial), and this is often very helpful. Many designers prefer a 100 thou visible grid and rarely vary from that.

Some programs also have what is called an “Electrical” grid. This grid is not visible, but it makes your cursor “snap” onto the center of electrical objects like tracks and pads, when your cursor gets close enough. This is extremely useful for manual routing, editing and moving objects.

One last type of grid is the “Component” grid. This works the same as the snap grid, but it’s for component movement only. This allows you to align components up to a different grid. Make sure you make it a multiple of your Snap grid.

When you start laying out your first board, snap grids can feel a bit “funny”, with your cursor only being able to be moved in steps. Unlike normal paint type packages which everyone is familiar with. But it’s easy to get used to, and your PCB designs will be one step closer to being neat and professional.

### Working from the top

PCB design is always done looking from the top of your board, looking through the various layers as if they were transparent. This is how all the PCB packages work. The only time you will look at your board from the bottom is for manufacturing or checking purposes. This “through the board” method means that you will have to get used to reading text on the bottom layers as a mirror image, get used to it!

### Rats Nest

Your job of component placement will be made infinitely easier by having a “rats nest” display enabled. If there is one reason for going to the trouble of drawing up an accurate schematic and importing a netlist, this is surely it. For large designs, a rats nest display is essential.

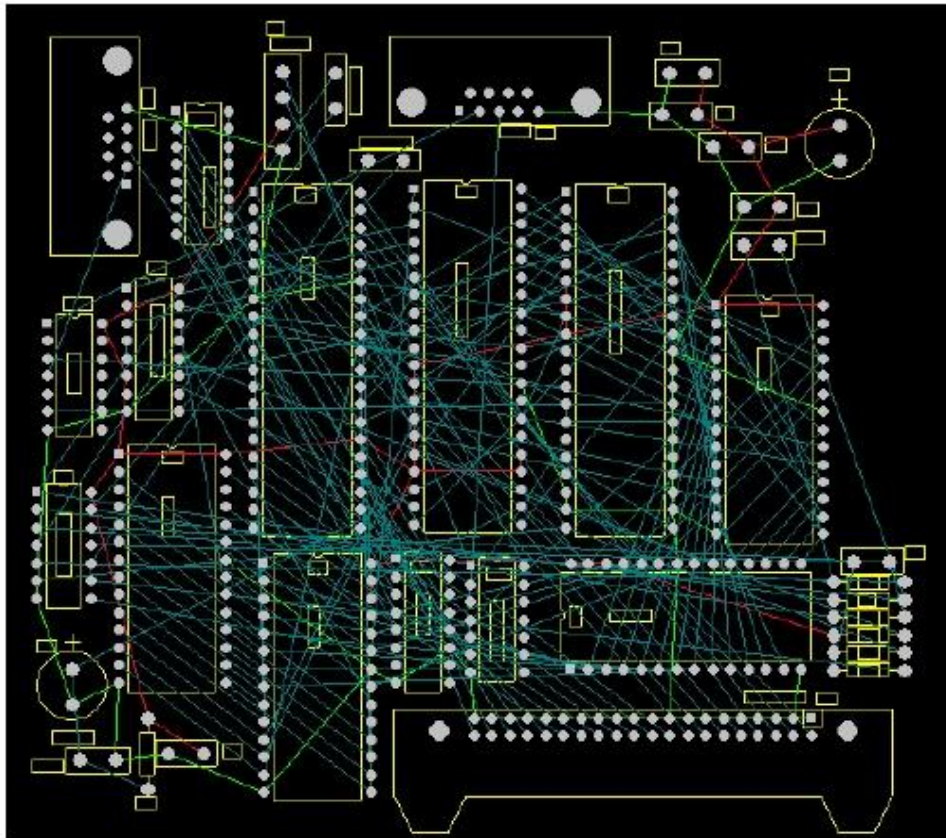
A rats nest display is one where the program will draw a straight line (not a track) between the pads of components which are connected on the schematic. In effect, it shows the connectivity of your circuit before you start laying out tracks. At the start of your board layout, with all your components placed down randomly, this will appear as a huge and complicated random maze of lines. Hence the name rats nest. The rats nest may look very daunting at first, but when you move each component the lines will automatically move with them. In this way you can see instantly which components are connected to which, without having to refer back to the schematic and constantly cross reference component designators. Once you have used this feature once, you won’t want to live without it. Even when doing simple designs with a few dozen components,

you will miss this functionality. With the rats nest display enabled, it will be almost possible to lay out all of your components optimally in no time, without having to lay down one single track. The rats nest display will effectively show you what your tracks will connect to. The rats nest lines should disappear when you route your tracks

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between components, so your design will get less and less “complicated looking” as you go along. When all the rats nest lines disappear, your board is fully routed.



*A Typical “Rats Nest” Display.*

### Tracks

There is no recommended standard for track sizes. What size track you use will depend upon (in order of importance) the electrical requirements of the design, the routing space and clearance you have available, and your own personal preference. Every design will have a different set of electrical requirements which can vary between tracks on the board. All but basic non-critical designs will require a mixture of track sizes. As a general rule though, the bigger the track width, the better. Bigger tracks have lower DC resistance, lower inductance, can be easier and cheaper for the manufacturer to etch, and are easier to inspect and rework. The lower limit of your track width will depend upon the “track/space” resolution that your PCB manufacturer is capable of. For example, a manufacturer may quote a 10/8 track/space figure. This means that tracks can be no less than 10 thou wide, and the spacing between tracks (or pads, or any part of the copper) can be no less than 8 thou. The

figures are almost always quoted in thou's, with track width first and then spacing. Real world typical figures are 10/10 and 8/8 for basic boards. The IPC standard recommends 4thou as being a lower limit. Once you get to 6thou tracks and below though, you are getting into the serious end of the business, and you should be consulting your board manufacturer first. The lower the track/space figure, the greater care the manufacturer has to take when aligning and etching the board. They will pass this cost onto you, so make sure that you don't go any lower than you need to. As a guide, with "home made" PCB manufacturing processes like laser printed transparencies and pre-coated photo resist boards, it is possible to easily get 10/10 and even 8/8 spacing.

Just because a manufacturer can achieve a certain track/spacing, it is no reason to "push the limits" with your design. Use as big a track/spacing as possible unless your design parameters call for something smaller. As a start, you may like to use say 25 thou for signal tracks, 50 thou for power and ground tracks, and 10-15 thou for going between IC and component pads. Some designers though like the "look" of smaller signal tracks like 10 or 15 thou, while others like all of their tracks to be big and "chunky". Good design practice is to keep tracks as big as possible, and then to change to a thinner track only when required to meet clearance requirements.

Changing your track from large to small and then back to large again is known as "necking", or "necking down". This is often required when you have to go between IC or component pads. This allows you to have nice big low impedance tracks, but still have the flexibility to route between tight spots. In practice, your track width will be dictated by the current flowing through it, and the maximum temperature rise of the track you are willing to tolerate. Remember that every track will have a certain amount of resistance, so the track will dissipate heat just like a resistor. The wider the track the lower the resistance. The thickness of the copper on your PCB will also play a part, as will any solder coating finish.

In high frequency designs PCB designer must take care to design "controlled impedance" traces, which match your electrical source and load impedance. It also allows you to keep signals coupled "tight" to their return path (ground). There are many ways to create controlled impedance "transmission" lines on a PCB. But the two most basic and popular ways are called Microstrip and Stripline. This topic is further explained in high frequency design technique.

The thickness of the copper on the PCB is nominally specified in ounces per square foot, with 1oz copper being the most common. You can order other thicknesses like 0.5oz, 2oz and 4oz. The thicker copper layers are useful for high current, high reliability designs.

The calculations to figure out a required track width based on the current and the maximum temperature rise are a little complex. They can also be quite inaccurate, as the standard is based on a set of non-linear graphs based on measured data from around half a century ago. These are still reproduced in the IPC standard.

A handy track width calculator program can be found at [www.ultracad.com/calc.htm](http://www.ultracad.com/calc.htm), and gives results based on the IPC graphs.

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As a rule of thumb, a 10degC temperature rise in your track is a nice safe limit to design around.

### Pads

Pad sizes, shapes and dimensions will depend not only upon the component you are using, but also the manufacturing process used to assemble the board, among other things. There are a whole slew of standards and theories behind pad sizes and layouts, and this will be explained later. Suffice it to say at this stage that your PCB package should come with a set of basic component libraries that will get you started. For all but the simplest boards though, you'll have to modify these basic components to suit your purpose. Over time you will build up your own library of components suitable for various requirements. There is an important parameter known as the pad/hole ratio. This is the ratio of the pad size to the hole size.

Each manufacturer will have their own minimum specification for this. As a simple rule of thumb, the pad should be at least 1.8 times the diameter of the hole, or at least 0.5mm larger. This is to allow for alignment tolerances on the drill and the artwork on top and bottom layers. This ratio gets more important the smaller the pad and hole become, and is particularly relevant to vias.

There are some common practices used when it comes to generic component pads. Pads for leaded components like resistors, capacitors and diodes should be round, with around 70 thou diameter being common. Dual In Line (DIL) components like IC's are better suited with oval shaped pads (60 thou high by 90-100 thou wide is common). Pin 1 of the chip should always be a different pad shape, usually rectangular, and with the same dimensions as the other pins.

Most surface mount components use rectangular pads, although surface mount SO package ICs should use oval pads. Again, with pin 1 being rectangular.

Other components that rely on pin numbering, like connectors and SIP resistor packs, should also follow the "rectangular pin 1" rule.

### Vias

Vias connect the tracks from one side of your board to another, by way of a hole in your board. On all but cheap home made and low end commercial prototypes, vias are made with electrically plated holes, called Plated Through Holes (PTH). Plated through holes allow electrical connection between different layers on your board. What is the difference between a via and a pad? Practically speaking there is no real difference, they are both just electrically plated holes. But there are differences when it comes to PCB design packages. Pads and Vias are, and should be, treated differently. You can globally edit them separately, and do some more advanced things to be discussed later. So don't use a pad in place of a via, and vice-versa. Holes in vias are usually a fair bit smaller than component pads, with 0.5-0.7mm being typical. Using a via to connect two layers is commonly called "stitching", as you are effectively electrically stitching both layers together, like threading a needle back and forth through material. Throw the term stitching a few times into a conversation and you'll really sound like a PCB professional!

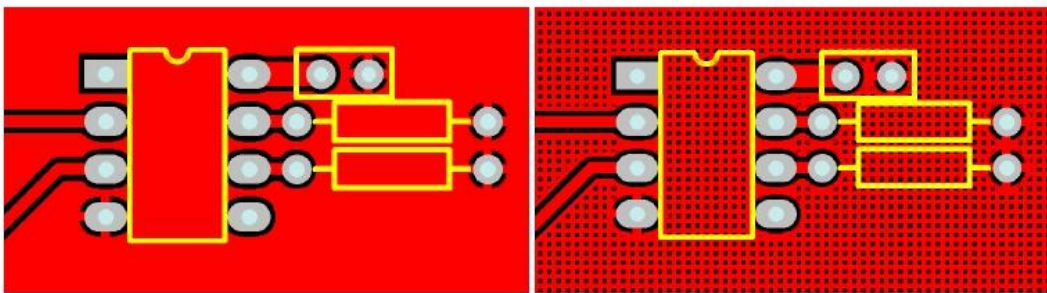


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### Polygons

“Polygons” are available on many PCB packages. A polygon automatically fills in (or “floods”) a desired area with copper, which “flows” around other pads and tracks. They are very useful for laying down ground planes. Make sure you place polygons after you have placed all of your tracks and pads. Polygon can either be “solid” fills of copper, or “hatched” copper tracks in a crisscross fashion. Solid fills are preferred, hatched fills are basically a thing of the past



An example of a "Solid Polygon Fill" (Left), and a "Hatched Polygon Fill" (Right)

### Clearances

Electrical clearances are an important requirement for all boards. Too tight a clearance between tracks and pads may lead to “hairline” shorts and other etching problems during the manufacturing process. These can be very hard to fault find once your board is assembled. Once again, don’t “push the limits” of your manufacturer unless you have to, stay above their recommended minimum spacing if at all possible. At least 15 thou is a good clearance limit for basic through hole designs, with 10 thou or 8 thou being used for more dense surface mount layouts. If you go below this, it’s a good idea to consult with your PCB manufacturer first.

For 240V mains on PCB’s there are various legal requirements, and you’ll need to consult the relevant standards if you are doing this sort of work. As a rule of thumb, an absolute minimum of 8mm (315 thou) spacing should be allowed between 240V tracks and isolated signal tracks. Good design practice would dictate that you would have much larger clearances than this anyway.

For non-mains voltages, the IPC standard has a set of tables that define the clearance required for various voltages. A simplified table is shown here. The clearance will vary depending on whether the tracks are on an internal layers or the external surface. They also vary with the operational height of the board above sea level, due to the thinning of the atmosphere at high altitudes. Conformal coating also improves these figures for a given clearance, and this is often used on military spec PCBs.

| Clearances for Electrical Conductors |          |                   |                   |
|--------------------------------------|----------|-------------------|-------------------|
| Voltage (DC or Peak AC)              | Internal | External (<3050m) | External (>3050m) |
| 0-15V                                | 0.05mm   | 0.1mm             | 0.1mm             |
| 16-30V                               | 0.05mm   | 0.1mm             | 0.1mm             |
| 31-50V                               | 0.1mm    | 0.6mm             | 0.6mm             |
| 51-100V                              | 0.1mm    | 0.6mm             | 1.5mm             |
| 101-150V                             | 0.2mm    | 0.6mm             | 3.2mm             |
| 151-170V                             | 0.2mm    | 1.25mm            | 3.2mm             |
| 171-250V                             | 0.2mm    | 1.25mm            | 6.4mm             |
| 251-300V                             | 0.2mm    | 1.25mm            | 12.5mm            |
| 301-500V                             | 0.25mm   | 2.5mm             | 12.5mm            |

Table 1.0

### Component Placement & Design

An old saying is that PCB design is 90% placement and 10% routing. Whilst the actual figures are of no importance, the concept that component placement is by far the most important aspect of laying out a board certainly holds true. Good component placement will make your layout job easier and give the best electrical performance. Bad component placement can turn your routing job into a nightmare and give poor electrical performance. It may even make your board unmanufacturable. So there is a lot to think about when placing components!

Every designer will have their own method of placing components, and if you gave the same circuit (no matter how simple) to 100 different experienced designers you'd get a 100 different PCB layouts every time. So there is no absolute right way to place your components. But there are quite a few basic rules which will help ease your routing, give you the best electrical performance, and simplify large and complex designs. At this point it is a good idea to give you an idea of the basic steps required to go about laying out a complete board:

- Set your snap grid, visible grid, and default track/pad sizes. Throw down all the components onto the board.

- Divide and place your components into functional "building blocks" where possible.

- Identify layout critical tracks on your circuit and route them first. Place and route each building block separately, off the board. Move completed building blocks into position on your main board. Route the remaining signal and power connections between blocks. Do a general "tidy up" of the board.

- Do a Design Rule Check.

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This is by no means a be-all and end-all check list, it's highly variable depending on many factors. But it is a good general guide to producing a professional first-class layout.

## Design Rule Checking

Design Rule Checking (DRC) allows you to automatically check your PCB design for connectivity, clearance, and other manufacturing errors. With the large and complex PCBs being designed today, it is impractical to manually check a PCB design. This is where the DRC comes into its own, it is an absolutely essential step in professional PCB design.

Examples of what you can check with a DRC are:

- Circuit connectivity. It checks that every track on your board matches the connectivity of your schematic.

- Electrical clearance. You can check the clearance between tracks, pads, and components

- Manufacturing tolerances like min/max hole sizes, track widths, via widths, annulus sizes, and short circuits.

A complete DRC is usually performed after you have finished your PCB. Some packages however have the ability to do “real time” (or “online”) DRC checking as you create your board. For instance, it won't let you connect a track to a pad it shouldn't go to, or violate a clearance between track and pad. If you have real-time DRC capability, use it, it's an invaluable tool.

## Forward and Back Annotation

Forward Annotation is when you make changes to your existing PCB layout via the schematic editor. The program will take your schematic netlist and component designators, and import them into your PCB design, and making any relevant changes. Some packages will also automatically remove old PCB tracks that are no longer connected. You can do this at any time during your PCB layout. If you update your schematic, then you must forward annotate into your PCB design. You can do edits like this manually, but forward annotation automates the process.

Back Annotation is when you change one of the component designators (eg. “C1” to “C2”) on your PCB and then automatically update this information back into your Schematic. More advanced back annotation features allow you to swap gates on chips, and perform other electrical changes. There should never be much real need to use back annotation.

## PCB design files to PCB Fabrication

After completing the PCB design we need to transfer our design files to PCB fabrication.

There are different PCB design tools available in market today from different vendors. The PCB design file output to PCB fabrication house normally contains a



group of files called Gerbers, NC drills and Fabrication drawing. All the PCB design tools are capable of generating these files from PCB design data base.

### High Frequency Design Techniques

High frequency design is where you really need to consider the effects of parasitic inductance, capacitance and impedance of your PCB layout. If your signal is too fast, and your track is too long, then the track can take on the properties of a transmission line. If you don't use proper transmission line techniques in these situations then you can start to get reflections and other signal integrity problems.

A "critical length" track is one in which the propagation time of the signal starts to get close to the length of the track. On standard FR4 copper boards, a signal will travel roughly 6 inches every nano second. A rule of thumb states that you need to get really concerned when your track length approaches half of this figure. But in reality it can actually be much less than this. Remember that digital square wave signals have a harmonic content, so a 100MHz square wave can actually have signal components extending into the GHz region. In high speed design, the ground plane is fundamental to preserving the integrity of your signals, and also reducing EMI emissions. It allows you to create "controlled impedance" traces, which match your electrical source and load. It also allows you to keep signals coupled "tight" to their return path (ground). There are many ways to create controlled impedance "transmission" lines on a PCB. But the two most basic and popular ways are called Microstrip and Stripline.

A Microstrip is simply a trace on the top layer, with a ground plane below. The calculation involved to find the characteristic impedance of a Microstrip is relatively complex. It is based on the width and thickness of the trace, the height above the ground plane, and the relative permittivity of the PCB material. This is why it is important to keep the ground plane as close as possible to (usually) the top layer.

A Stripline is similar to the Microstrip, but it has an additional ground plane on top of the trace. So in this case, the trace would have to be on one of the inner layers. The advantage of stripline over microstrip is that most of the EMI radiation will be contained within the ground planes.

There are many free programs and spreadsheets available that will calculate all the variations of Microstrip and Stripline for you.

### Design For Manufacturing

In general terms all defects found in manufacturing are caused by the design, the process or material. However, according to a study conducted to evaluate the impact of design on product quality, the cause of 40% of quality problems was found to be poor design. A product design must not only meet performance requirements but should also minimize life cycle cost and technical risks.

This is accomplished through an awareness of how design decisions affect the manufacturing process, including capabilities and limitations. DFX is a philosophy of designing a product so that it can be produced with the least cost and highest level of quality and reliability. DFX is both an analytical process and a creative science that draws on many engineering disciplines. The most important single design practice for

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Manufacturability is simplicity. A simple design can be manufactured at higher levels of quality and lower cost than a complex design. A critical factor in developing designs that are easy to produce is a clear understanding of the manufacturing process capabilities and limitations. Design requirements and guidelines must incorporate small changes in design parameters and tolerances that facilitate ease of operation for a particular manufacturing process.

## Panelisation

If you are looking at getting your board automatically assembled with a pick-and-place machine, then it pays you to get as many boards onto the one “panel” as you can. A panel is simply a large PCB containing many identical copies of your board. It takes time to place a board into position on a pick and place machine, so the more boards you can load at once, the more cost effective your manufacturing will be. A panel will also contain tooling strips on the top and bottom, to allow for automated handling of the panel. Different manufactures may have different maximum panel sizes they can produce. Each individual board can be “routed out” and joined with “breakout tabs”, or simply butted together and scoured out with a “V groove”. A V groove is a score mark placed on your board that allows you to easily “snap” the board along the groove. A breakout tab is a small strip of board perhaps 5-10mm long joining your board to your panel. Small non-plated holes are also drilled along this strip, which allows the board to be snapped or cut out of the panel after assembly. You will need to consult your board loader to determine optimum panelisation size and requirements.

## Tooling Strips

Tooling strips are strips of blank board down the top and bottom side of your board. They contain tooling holes, fiducial marks, and other manufacturing information if required.

Standard tooling holes are required for automated handling of your board. 2.4mm and 3.2mm are two standard hole sizes. Four tooling holes per panel is sufficient, one in each corner.

The tooling trips connect to your board(s) with breakout tabs or V Grooves.

## Fiducial Marks

Fiducial marks are visual alignment aids placed on your PCB. They are used by automated pick and place machines to align your board and find reference points. A video camera on the machine can identify the center of fiducial marks and use these points as a reference. On a panel there should be 3 fiducial marks, known as global fiducials. Bottom left/right and top left corners. They should be at least 5mm away from the board edges. They can be mounted on the tooling strips. The fiducial mark should be a circular pad on the copper layer of diameter 1.5mm typically. The fiducial should not be covered with solder mask, and the mask should be removed for a clearance of at least 3mm around. The pad can be bare copper or coated like a regular pad. Two local fiducial (one in opposite corners) is also required next to each large fine pitch surface mount device package on your board.

### Thermal Relief

If you solidly connect a surface mount pad to a large copper area, the copper area will act as a very effective heat sink. This will conduct heat away from your pad while soldering. This can encourage dry joints and other soldering related problems. In these situations a thermal relief connection, which comprises several (usually 4) smaller tracks connecting the pad to the copper plane. Thermal relief options can be set automatically in many packages.

### Soldering

Soldering considerations need to be taken into account when laying out your board. There are three basic soldering techniques - hand, wave, and reflow. Hand soldering is the traditional method typically used for prototypes and small production runs. Major impacts when laying out your board include suitable access for the iron, and thermal relief for pads. Non-plated through double sided boards should allow for ample room to get the soldering iron onto the top side pads. Wave soldering is a common process used for surface mount and through hole production soldering. It involves passing the entire board over a molten bath of solder. Solder masks are absolutely essential here to prevent bridging. The major thing to watch out for when designing is ensuring that small components are not in the wave solder “shadow” of larger components. The board travels through the wave solder machine in one direction, so there will be a lack of solder trailing behind larger components. Surface mount devices are fixed to the board with an adhesive before wave soldering.

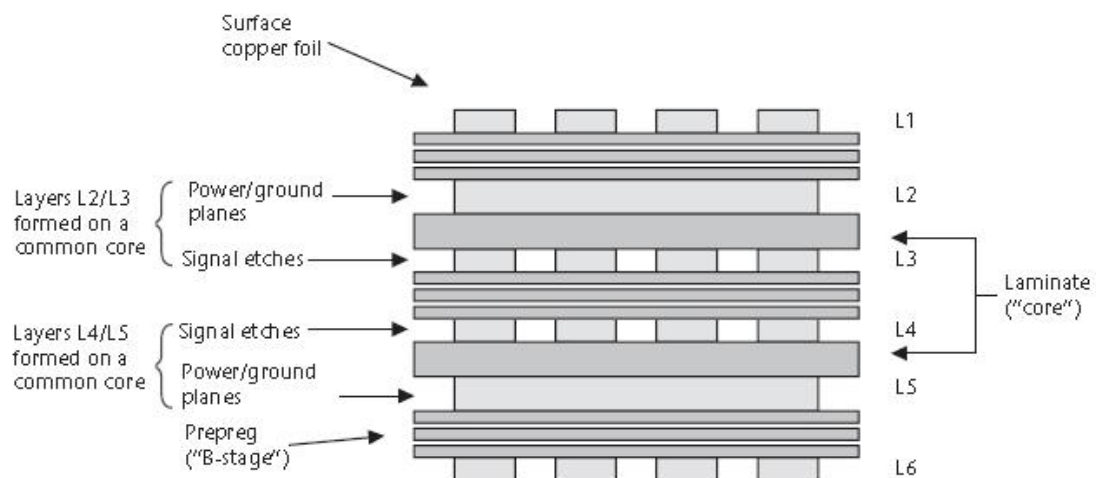
Reflow soldering is the latest technique, and is suitable for all surface mount components. The blank board is first coated with a mask of solder paste over the pads (solder “stencils” are used for this). Then each component is placed, and is sometimes held in place by an adhesive. The entire board is then loaded into an infrared or nitrogen oven and “baked”. The solder paste melts (reflows) on the pads and component leads to make the joint.

A newer reflow method called pin-in-paste or intrusive reflow is available for through hole devices. Combinations of wave and reflow soldering can be used for mixed through hole and surface mount boards. Wave soldering has the advantage of being cheap, but the disadvantage of imposing placement limits on your components. Reflow soldering is more complex and expensive, but it allows for very dense surface mount packing.

## Multilayer PCB Construction

The typical multilayer PCB is formed as a stack of alternating layers of prepreg mats and laminate sheets. The general idea is shown in Figure 1.1.

The prepreg mats are a weave of glass fiber yarns preimpregnated (hence *prepreg*) with a resin that is intentionally allowed to only partially cure. The sheets come in many stock sizes and yarn *styles* (classified by the number and diameter of the glass threads, the weave, and the percentage of resin impregnation) and serve to strengthen the resin. The typical resin content of the mats is in the 45% to 65% range.



**Figure 1.1** Multilayer PWB stackup.

Copper foil is attached to one or both sides of fully cured prepreg sheets to form the laminate sheets (also called *cores*). Similar to the prepreg mats, cores come in standard stock sizes and thicknesses, from which the fabricator must choose to construct a PCB. It's common for outer layers (such as layers L1 and L6 in Figure 1.1) to be formed on prepreg, but some manufacturers prefer to form the outer layers on cores.

To form the composite PCB structure, a stackup of prepreg mats and laminate cores are heated under pressure. This causes the partially cured prepreg to flow and bond to the cores. The prepreg cures are cooled, thereby forming the completed PCB structure.

## Resins

Several resin systems are used to form prepreg and laminate sheets, with the FR4 epoxy resin system probably being the most popular. The generic specification *FR4* refers to a specific fire-retardant level rather than to a specific laminate chemistry. The term *standard FR4* is a misnomer: Fabricators have many laminate systems from which to choose that meet FR4 fire specifications, each with unique electrical and mechanical characteristics. Fabricators select the laminate system based on assembly issues (especially temperature extremes) and their history with a particular laminate for a given set of electrical requirements. The FR4 epoxy can be blended with tetrafunctional or multifunctional resins to improve the material's mechanical characteristics, most notably the coefficient of expansion, the *glass transition temperature*, and the rate of moisture absorption.

The coefficient of expansion in height—that is, the *Z* axis, *CTE(Z)*—is an indication of how much the board will enlarge and contract in thickness with temperature changes. Increasing the resin's temperature above its glass transition temperature (*T<sub>g</sub>*) causes the resin to change from its crystalline state to a more fluid, glassy state. Additionally, *CTE(Z)* is not constant across temperature: increasing temperature above *T<sub>g</sub>* results in a rapid increase in *CTE(Z)*. Fabricators will recommend the use of a *high T<sub>g</sub>* laminate (those with *T<sub>g</sub>* greater than about 180°C) if the PCB will be exposed to high temperatures during assembly or rework operations. This is important because thermally induced stresses can lead to plated through hole failures, especially in thick boards having a large *CTE(Z)*.

The drilling operation is another consideration for the fabricator when choosing a laminate. The typical FR4 resin system is relatively easy to drill. High *T<sub>g</sub>* laminates tend to be harder and more brittle, making drilling more difficult. On the other hand, high-speed drilling of vias may warrant the use of a high *T<sub>g</sub>* material to avoid *drill smear*. This is a result of a softening of the laminate near the hole due to drilling induced local heating that momentarily raises the temperature above *T<sub>g</sub>* (*T<sub>g</sub>* 130°C for FR4).

## Alternate Resin Systems

Resins other than the standard FR4 epoxy types are available to form PCB laminates and prepregs. These alternate systems have higher *T<sub>g</sub>* than the tetrafunctional or multifunctional FR4 resins, and superior electrical characteristics. Of principal interest are the values for the dielectric constant.

As shown, the multifunctional FR4 epoxies have the lowest *T<sub>g</sub>* and highest  $\epsilon_r$ /loss tangent values. Of the resins, polyamide has the highest *T<sub>g</sub>* and  $\epsilon_r$ /loss tangent values, second to the cyanate ester resin system. The Rogers RO4350® has the highest *T<sub>g</sub>* and lowest  $\epsilon_r$  and loss tangent value.

**Table 1.1** Alternate Laminate Systems

| Trade Name | Chemistry       | Tg   | $\epsilon_r$ /loss tan | $\epsilon_r$ /loss tan | Vendor     |
|------------|-----------------|------|------------------------|------------------------|------------|
|            |                 | C°   | 1 MHz                  | 1 GHz                  |            |
| N7000-1    | Polyamide       | 260  | 4.3/0.013              | 3.7/0.007              | Nelco      |
| P97        | Polyamide       | 260  | 4.4/0.014              | 4.2/0.014              | Isola      |
| N8000      | Cyanate Ester   | 250  | 3.8/0.008              | 3.5/0.006              | Nelco      |
| N5000      | BT              | 185  | 4.1/0.013              | 3.8/0.010              | Nelco      |
| G200       | BT              | 185  | 4.1/0.013              | 3.9/0.009              | Isola      |
| N4000-6    | Multifunctional | 180  | 4.4/0.023              | 3.9/0.012              | Nelco      |
| Megtron    | PPO/Hi Tg Epoxy | 180  | 3.8/0.010              | 3.75/0.011             | Matsushita |
| FR404      | Multifunctional | 150  | 4.6/0.025              | 4.25/0.014             | Isola      |
| RO4350     | Ceramic         | >280 |                        | 3.48/0.004<br>(10GHz)  | Rogers     |

### Reinforcements

Fibers (usually from a form of glass) are used to strengthen the resins, but adding them changes the electrical and mechanical characteristics of the composite structures roughly in proportion to the amount of fiber to resin (the *glass-to-resin ratio*). As shown in Table 1.2, the glass fibers have a higher  $\epsilon_r$  but superior loss tangent values than the resins.

High glass content improves the composite's CTE(Z), thereby helping to prevent via cracking during high-temperature assembly and rework operations. However, a high glass content increases  $\epsilon_r$  and lowers the loss tangent. Generally, higher  $\epsilon_r$  is a disadvantage in high-speed PCBs, as that increases capacitive coupling between conductors and tends to result in thicker stackups for a given impedance. Alternatively, for a given stackup thickness, higher  $\epsilon_r$  results in narrower trace widths, thereby increasing conductor loss.

Various glass fiber types are available to reinforce the resin. The most common is *E* glass (electrical grade), which is commonly used throughout the plastics industry. This glass fiber was specifically designed for electrical use, but its versatility has made it suitable for reinforcing a range of plastics. This broad adoption beyond use in the PCB industry is responsible for the low cost of *E* glass. It primarily consists of silicon oxide, aluminum oxide, and calcium oxide.

A second glass fiber finding increasing use in PCBs is *S* glass (structural grade). This glass fiber was specifically developed for high-strength reinforcement applications

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and also consists of silicon oxide and aluminum oxide, but it uses magnesium oxide in place of the calcium oxide. It's stronger than E glass and has a lower  $\epsilon_r$ , but it's not as widely used and so is more expensive (about four times that of E glass)

**Table 1.2** Resin and Reinforcement Properties at 1 MHz

| <i>Material</i>     | $\epsilon_r$ | <i>Loss</i> | <i>CTE(Z)</i><br><i>Parts Per Million</i><br><i>(PPM)/C°</i> | <i>Moisture</i><br><i>Absorption</i><br><br>(%) |
|---------------------|--------------|-------------|--------------------------------------------------------------|-------------------------------------------------|
| E Glass             | 6.2          | 0.004       | 5.5                                                          |                                                 |
| S Glass             | 5.2          | 0.003       | 2.6                                                          |                                                 |
| Thermount®          | 3.9          | 0.015       | -4.5                                                         | 0.44                                            |
| FR4 epoxy resin     | 3.6          | 0.032       | 85                                                           | 0.7                                             |
| BT resin            | 3.1          | 0.003       |                                                              |                                                 |
| Polyamide resin     | 3.2          | 0.02        | 50                                                           | 0.9                                             |
| Cyanate Ester resin | 2.8          | 0.002       | 50                                                           | 0.5                                             |

Source: [4, 7, 15–20].

### Variability in Building Stackups

In producing a multilayer PCB as depicted in Figure 1.1, the fabricator must decide on the thickness of the laminate and the styles, thickness, and the number of prepreg mats to use to form each layer. The laminate sheets tend to have lower resin content than the prepreg, so the laminates usually have higher  $\epsilon_r$  than the prepreg sheets. The way in which the fabricator chooses to form the stackup is fundamental in that it determines  $\epsilon_r$  and the loss tangent for a particular layer. One fabricator may choose to use several thin, high-resin-content mats resulting in a lower overall  $\epsilon_r$ , while another prefers to use a single, thicker mat having lower resin content which will yield a higher  $\epsilon_r$  to get the same overall thickness.

The vendor's latitude in making all of these choices means that ostensibly identical PCBs fabricated by different vendors will quite naturally have different electrical properties.

### Mixing Laminate Types

It's not necessary for the cores to all be the same laminate material. Historically this hybrid-type construction has been expensive and not widely used in the commercial digital PCB industry, but in recent years it's become somewhat more mainstream. In spite of its fabrication complexity, it can be cost effective to use a lower cost, lower performance laminate throughout most of the stackup and strategically mix in a few expensive, higher performance layers only where needed. This is especially attractive

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for stackups that have many layers, where only a few carry high frequency, loss sensitive signals.

In these situations, the complexity of fabricating a stackup containing different materials costs less than making the high layer count stackup entirely from the high-performance, expensive laminate. The materials chosen must have similar CTE values. So not all laminate types can be mixed. Also note that the copper thickness need not be the same throughout the stackup. Having different copper thickness on various layers is common in situations where the power/ground planes must be thick for proper power supply distribution but the signal traces need not be.

At high frequency the skin effect causes signal currents to migrate to the conductor's surface. This means thick traces do not necessarily have a loss advantage over thinner ones. An additional advantage of using thin copper for signal traces is that it's easier to retain a truly rectangular shape when etching the thinner copper. This has loss, coupling, and impedance advantages. Using thinner copper for the signal traces can help reduce the stackup's thickness, but using thin copper for the power/ground planes reduces their ability to wick heat from the pins of an integrated circuit or field effect transistor (FET) (as is found in switching power supplies or when FETs are used in power supply sequencing circuits).

To avoid warpage, manufacturers favor *balanced* stackups, where the thin and thick layers are distributed symmetrically about the stackup's center, but this also applies to the distribution of laminate types.

### PCB Traces

Copper traces are used to form the PCB conductors, either of the board's surface (*microstrip* or *embedded microstrip*) or buried within the PCB as *stripline* (see Figure 1.2).

The nature of the multilayer PCB fabrication process is such that the mechanical characteristics of the inner layer copper is different from the copper on the board's outer surfaces. The outer layers are plated, while the inner ones are not.

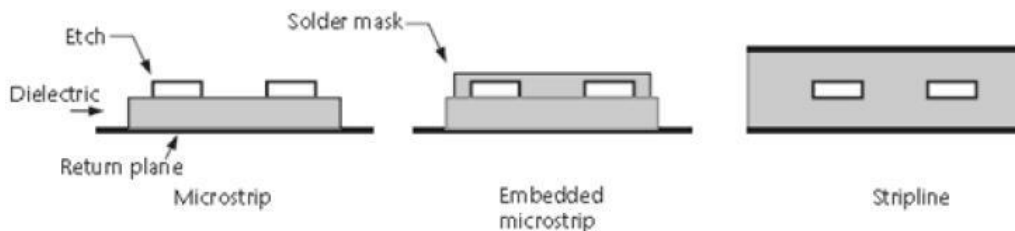


Figure 1.2 Micro strip and strip line defin.

### Copper Cladding

The copper cladding attached to laminate sheets is created by either an electrodeposition or rolling process. These processes create copper foils with different surface roughness. Accounting for surface roughness is important when computing



skin effect losses. The electrodeposited process (ED) creates copper foil by a plating process that forms a copper sheet by extracting copper from solution onto a rotating drum.

The foil side in contact with the drum is smoother than the other surface. In contrast, the rolling process starts with a copper ingot that is passed through rollers multiple times until it is reduced to the desired thickness. This process creates foil equally smooth on both sides, and smoother than that of electrodeposited copper. A smooth surface is advantageous when signaling at high frequency because the ac resistance will be lower with a smooth surface than it will be with a rough one. This makes rolled copper trace electrically preferred over ED at high frequencies.

However, the greater coarseness of ED foil allows the copper to better adhere to the substrate, giving ED foils higher *peel strengths*. Foils with higher peel strengths have better adhesion and so are less likely to lift off from the laminate during soldering or rework operations.

To promote adhesion with the laminate material, both types of foils are roughened on one side (or sometimes both sides) to increase surface area. There are many techniques available for fabricators and laminators to use, each producing different copper grain sizes and shapes. Surface roughness is measured as the root mean-square (RMS) height of the irregularity above the surface.

As shown in Table 1.3, in general, even after processing, rolled copper has a lower surface roughness than ED. The data in Table 1.3 should only be taken as representative. Actual values depend on processing and will vary between manufacturers. Because the CTE of copper foil is actually lower than that of the laminate, thermally induced stresses can cause the connection to a via to fracture over time or with repeated thermal cycling. High-temperature elongation (HTE) foils can be used to mitigate this susceptibility to stress. These foils are also sometimes called *class 3 foils* after the Institute for Interconnecting and Packaging Electronic Circuits (IPC) industry standards group designation. Foils in this category have a higher CTE than the *class 1 foils* that are in general use and more closely match the laminate's CTE. The use of HTE foils is becoming common, especially on higher performance resin-based laminates, but they are generally not used with laminate systems having a low CTE(Z), such as the Rogers 4000® series materials.

**Table 1.3** Typical Copper Foil Characteristics

|            | Average Thickness<br>(mils) | ED $\mu$ -inches<br>(RMS) | Rolled (Treated Side)<br>$\mu$ -inches (RMS) |
|------------|-----------------------------|---------------------------|----------------------------------------------|
| Half ounce | 0.65                        | 75–100                    | 50–60                                        |
| One ounce  | 1.4                         | 95                        | 50–60                                        |

Source: [24, 25].

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### Copper Weights and Thickness

The thickness of the copper foil is usually specified by its nominal weight in ounces per square foot of area. Table 1.4 shows the relationship of weight to nominal and minimum thickness as specified by the IPC.

Notice that due to plating, the external conductors (i.e., microstrip) will usually be thicker than the inner layers (stripline) of the same weight.

**Table 1.4** Relationship Between Copper Weight Specification and Thickness

| <i>Weight Specifier<br/>(oz)</i> | <i>Nominal Thickness<br/>(mils)</i> | <i>Minimum Internal<br/>Layer Thickness (mils)</i> | <i>Minimum External<br/>Layer Thickness (mils)</i> |
|----------------------------------|-------------------------------------|----------------------------------------------------|----------------------------------------------------|
| $\frac{1}{4}$                    | 0.35                                | 0.25                                               | 0.8                                                |
| $\frac{1}{2}$                    | 0.70                                | 0.50                                               | 1.30                                               |
| 1                                | 1.4                                 | 1.0                                                | 1.8                                                |
| 2                                | 2.8                                 | 2.2                                                | 3.0                                                |

### Vias

Vias are used to connect traces appearing on different layers. A stackup with vias connecting L1 to L3 and another connecting L3 to L4 is shown in Figure 1.3. Vias are formed by drilling (either mechanically or with a laser) a hole partially (*blind vias*) or completely through the PCB stackup (*through hole vias*). Those vias that are to be made conductive are then plated. Conductive holes passing entirely through the PCB are called *plated through holes* (PTH).

An annular ring of copper (called a *land* or *pad*) surrounds the via to insure that even with some layer-to-layer misalignment, the drilled hole can still contact the trace on each of the required layers. An off-center via is still able to make complete electrical contact with the copper trace..

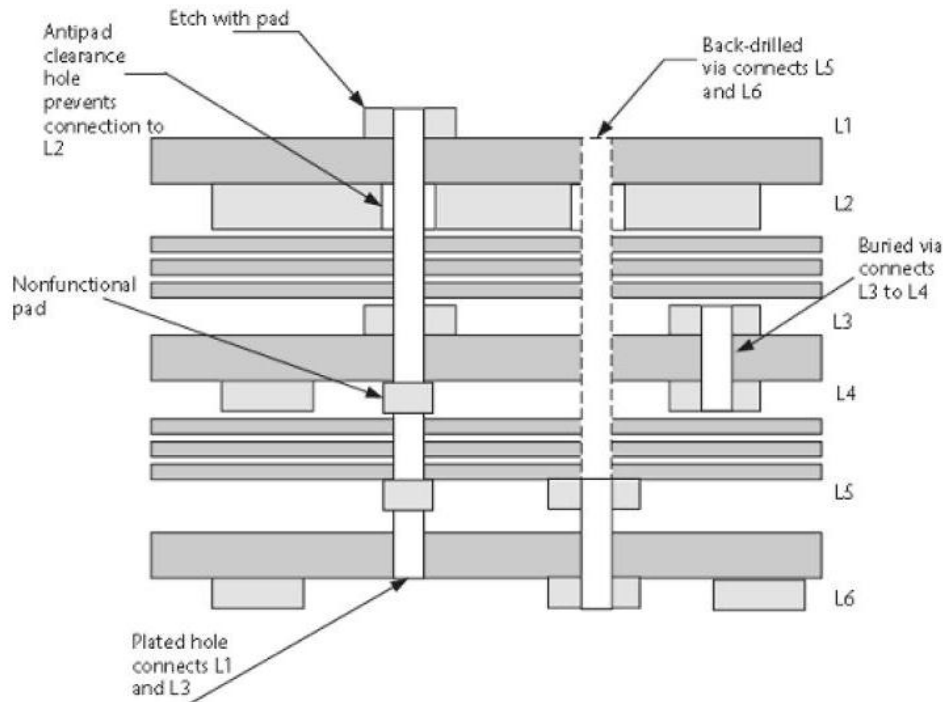


Figure 1.3 Vias connecting L1 to L3 and L3 to L4

Figure 1.3 Vias connecting L1 to L3 and L3 to L4

These pads are usually only placed on the layers where the via is to connect to a trace, but they can also be placed on the via at where the via does not connect to a trace. These *nonfunctional pads* are visible in Figure 1.4 and serve to anchor the via in the stackup. They are more common on thick boards because vertical stresses caused by thermal expansion are generally greater there than on thin PCB having only a few layers. These nonfunctional pads are often electrically benign, but they do increase a via's self capacitance, which can be detrimental in low-jitter, high-frequency systems.

A clearance hole (sometimes called an *antipad*) is made in the power and ground planes when it's necessary for a via to pass through the plane without making contact.

The size of this antipad is an important factor in determining the amount of capacitive coupling that occurs from the via to the plane. This is an important consideration in high-speed interconnect. The antipad construction appears in Figure 1.4

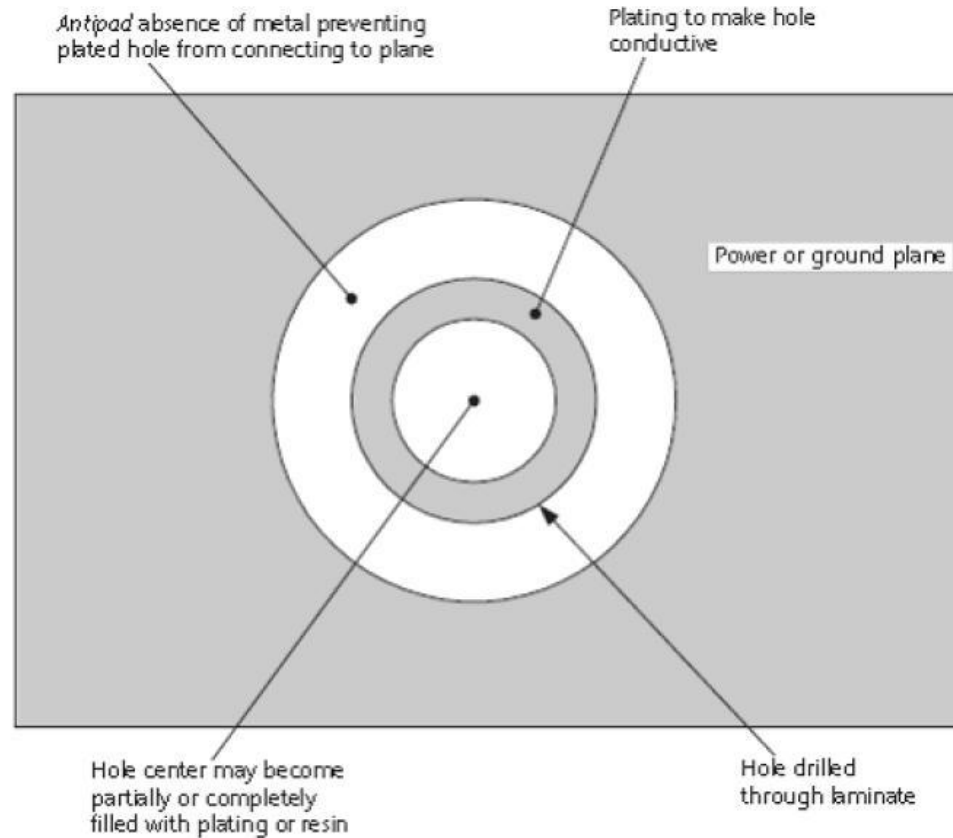


Figure 1.4 Antipad construction

A *thermal relief* pad (diagrammed in Figure 1.5) is placed on the plane in those situations when the via is to connect to the plane. This helps to thermally isolate the via from the plane, preventing the plane from acting as a heat sink and wicking away heat from the via during soldering. Notice that the connection is made by four tabs from the via body to the plane, increasing the via's resistance and inductance.

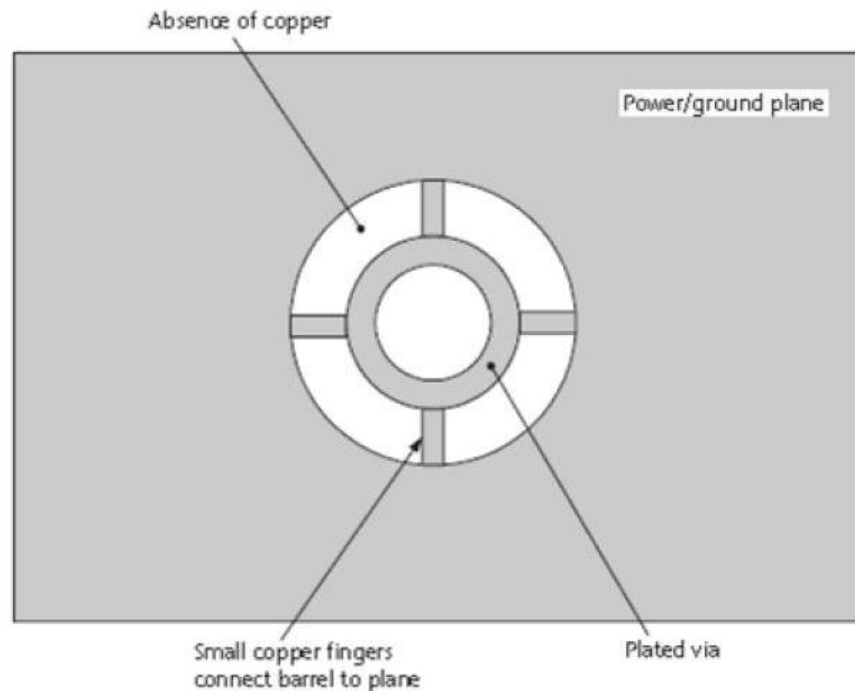


Figure 1.5 Thermal relief via

### Via Aspect Ratio

Vias that are too long relative to their diameter make the PCB difficult to manufacture and can pose long-term reliability problems. The ratio of the PCB thickness to a via's outside diameter is a commonly used metric called the *via aspect ratio* (or sometimes simply the *aspect ratio*).

$$\text{Aspect Ratio} = \frac{\text{PCB thickness}}{\text{Via O.D.}}$$

Fabricators use the via aspect ratio as one way to categorize a PCB's complexity to be manufactured. Each fabricator will set their own aspect ratio limits based on their technical prowess and historical ability. It's generally easy to find fabricators willing to manufacture low-aspect ratio boards, but as the aspect ratio increases the cost per board goes up and the number of capable shops decreases. Low-aspect ratio boards are thus easy to second source and will cost less than those with high-aspect ratios.

Low-aspect ratios (currently considered to be under about eight) are generally the easiest to produce; at the moment, aspect ratios from eight to roughly 10 are considered mainstream and ratios greater than 10 are presently considered to be in the difficult or advanced category. At a significantly higher cost, some advanced fabricators can reliably manufacture boards with aspect ratios approaching 14. This is only a guide; the categorization varies between manufacturers and is somewhat dependent on laminate type (especially for the highest aspect ratios).

## Surface Finishes and Solder Mask

Traces appearing on the boards' surface are protected from corrosion by the application of a *surface finish*. Elimination of corrosion aids in soldering and improves the reliability of soldered connections. Many surface finishes are available, and the correct choice for a particular situation is determined by such factors as the amount and type of rework the board will be subjected to, the degree of surface coplanarity required, and the shelf life (time the fabricated bare boards can wait before they are assembled).

Some of the more common surface finishes include hot air solder leveling (HASL), which is a coating of solder that can vary significantly in thickness across the board; organic solderability preservative (OSP), which is a very thin application of an organic material; electroless nickel immersion gold (ENIG); or immersion silver or immersion tin.

One would expect that the choice of surface finish would impact high-frequency conductor losses, and indeed this is so. The highest losses have been reported (in descending order) with immersion gold, followed by immersion tin, OSP, and immersion silver having the lowest losses.

*Solder mask* is another coating that is applied to a board's surface. This is sometimes confused with surface finish, but the two are quite distinct. Where surface finish coats all exposed copper to prevent corrosion, solder mask is a thin coating of epoxy placed everywhere on the board except where electrical connections are to be made (such as component solder pads and gold-plated fingers for edge connectors).

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